GMI German-Malaysian Institute

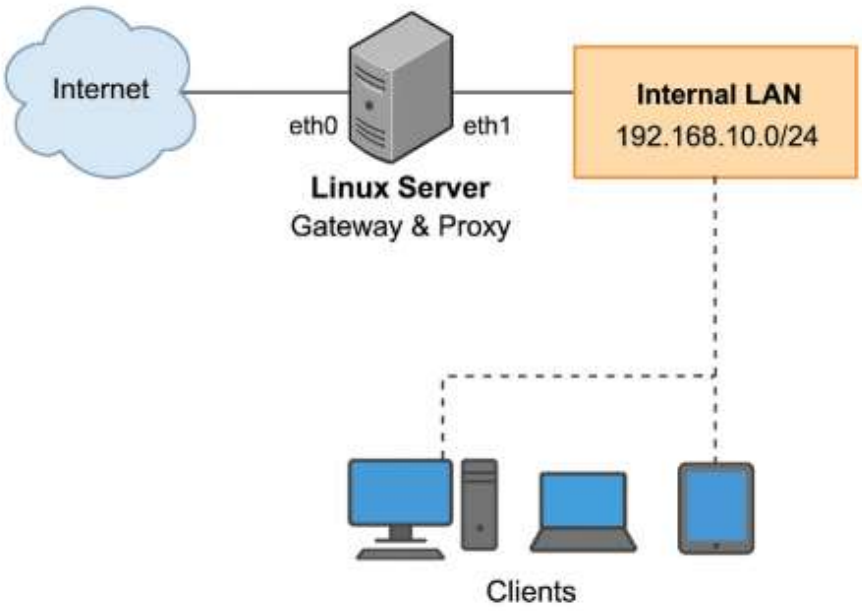
DIPLOMA PROGRAMME

MINI PROJECT 30%

Academic Period: JAN 2026

Code	CBS 2413	Course Name	SECURING NETWORK
Title	WEB PROXY WITH SQUID		
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Group	CBS 3	Examiner	MOHD HAFIZ RAZALI

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	All criteria are based on the rubric provided.	2, 3	2, 6	P3, A3	32			
2								
3.								
4.								
5								
				Total		[redacted]	[redacted]	[redacted]
				Total Marks	/32			
		Total	PO		/32 X 30%			
				Signature				



QUESTION:

In this project, you are required to design and implement a secure network using iptables and squid proxy that controls and monitors internet access for a small office network. Give the solution includes traffic filtering, access control, and to optimize performance and security.

PROJECT REQUIREMENTS:

Virtualized Network Design:

- Create a virtual network topology within ESXi, simulating a small office or lab environment. This should include at least two LANs representing different network segments – internal and external.
- Document the network design, including IP addressing schemes, LAN configurations, and logical network diagram.

Iptables configuration:

- Allow only HTTP traffic from LAN.
- Block access to specific external websites or IP addresses.

Squid Proxy setup:

- Block access to social media and entertainment websites

Testing and Verification:

- Develop test cases to verify the functionality of the security policies.
- Document the test results.

Documentation:

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- Create a project video, including:
    - Network design documentation.
    - Step-by-step configuration instructions.
    - Test results and analysis.
    - Lessons learned and recommendations for future improvements. Include group members at the wallpaper of the video.

Deliverables:

- Virtual network topology diagram.
- Test plan and results.
- Project documentation.

Rules and Regulations:

- Each group may have a maximum of 4 members (permitted to be done individually)
- Each group must complete the project by the assigned deadline within timeline given.
- All group members must actively participate and contribute to the project.
- Ensure that all sources used in the project are properly cited.
- Each group must present their findings clearly (narration and subtitle)
- Late submissions will be penalized according to the project guidelines.

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RUBRICS: SECURING A VIRTUALIZED NETWORK ENVIRONMENT WITH SQUID WEB PROXY

NO.	CRITERIA	EXCELLENT (4)	GOOD (3)	FAIR (2)	NEED IMPROVEMENT (1)	SCORE
	GROUP COMPONENT (TOTAL MARK = 24)					
1	Network Design & Implementation	Flawless design, accurate config.	Minor errors, mostly correct	Noticeable errors, partially correct.	Major errors, incomplete.	/4
2	Squid Proxy Configuration	Perfect deployment, optimized config.	Minor errors, functional.	Noticeable errors, unstable.	Incomplete, non-functional.	/4
3	Security Policy Implementation	Effective, accurate, optimized policies.	Minor errors, functional policies.	Noticeable errors, partial policies.	Incomplete or incorrect policies.	/4
4	Testing & Verification	Comprehensive test plan, thorough testing.	Basic testing, present results.	Incomplete plan, limited testing.	No testing, missing results.	/4
5	Content & Presentation	Clear, well-organized content, professional.	Clear and adequate presentation.	Unclear, poor presentation.	Missing/poor video, no presentation.	/4
6	Group Collaboration	Equal contribution, excellent teamwork.	Mostly good contribution, teamwork.	Uneven contribution, fair teamwork.	Imbalance, poor teamwork.	/4
	INDIVIDUAL COMPONENT (TOTAL GRADE = 8)					
7	Individual Contribution	Detailed, reflective, insightful.	Present, outlines contributions.	Incomplete, vague.	Missing, no description.	/4
8	Individual Technical	Deep understanding, effective troubleshooting.	Good understanding, basic troubleshooting.	Fair understanding, struggles.	Poor understanding, no troubleshooting.	/4
					TOTAL	**/32**

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1.0 INTRODUCTION

1.1 Purpose of the project

The internet can be a problem in a workplace. If people can use the internet without any limits, it can make the workplace computers and network vulnerable to people and viruses. This project is about making a security network for a small office. We want to use a kind of computer server that can control what happens when the office computers talk to the outside world.

The main thing we want to do is make a system that can watch what people are doing on the internet and block things that are not allowed. We will use something called iptables and a Squid web proxy to do this. These tools will help keep the office network from bad people on the outside and make sure people are following the company rules about using the internet. The internet is a part of the workplace, and the company's internet usage policies must be enforced to keep everything safe and secure.

We will use the iptables and the Squid web proxy to monitor the traffic on the network and filter the web content. This way the internal network will be protected from threats and the company rules will be followed. The internet access in the workplace will be safe and secure, with these tools.

1.2 Background

Network administrators frequently struggle to balance open web access with tight data security. Traditional firewalls operate mainly at the network and transport layers, meaning they excel at blocking specific ports or IP addresses but fail to understand the actual content passing through them. This is where a web proxy becomes necessary.

Squid is a caching proxy that operates at the application layer of the OSI model. It intercepts web requests from clients, checks them against predefined access control lists, and either allows or drops the request. It also catches frequently visited pages, which helps save local bandwidth and speeds up load times for users. Pairing Squid with iptables, the native packet-filtering framework in Linux creates a defensive system that secures both the network routing level and the web browsing level.

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1.3 Scope and Focus

This project focuses on building a virtualized small office network using an ESXi environment. The network topology consists of two distinct segments: an untrusted external network representing the Internet, and a private internal Local Area Network (LAN) using the 192.168.10.0/24 subnet.

The security policies implemented are limited to:

- Restricting outbound LAN traffic through iptables to allow only essential web and DNS services.
- Creating application-layer rules in Squid to block access to specific social media platforms and entertainment websites.
- Blocking specific file extensions to prevent users from downloading unauthorized media or potentially malicious executables.

  Advanced enterprise deployments, such as full SSL inspection (decryption) or integration with external directory services (like Active Directory), fall outside the scope of this assignment.

1.4 Methodology

The practical phase of this deployment relies on a specific software and virtualization matrix:

- VMware ESXi Hypervisor: Used to host and run our virtual systems while providing isolated virtual switches to separate local test traffic from the external world.
- Linux Gateway Appliance: An Ubuntu Server VM functioning as our central router and security boundary, running dual virtual interfaces (ens33 for internet routing and ens37 for local network hosting).
- Linux Client Instance: A Kali Linux desktop virtual machine configured to communicate solely within our private subnet to test and verify firewall behavior.
- Security Tools: Native Linux iptables for packet masquerading and port control, and a standard Squid daemon for domain restriction rules.

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1.5 Significance of the project

Commercial enterprise firewalls are often far too expensive for startups or smaller branch offices. This assignment highlights how simple, free, open-source software solutions can achieve high-level protection on minimal hardware budgets. Working directly with configuration structures in iptables and Squid provides essential system administration experience, teaching practical skills in routing control, policy enforcement, and infrastructure design.

2.0 CONTENTS

2.1 Network design and implementation

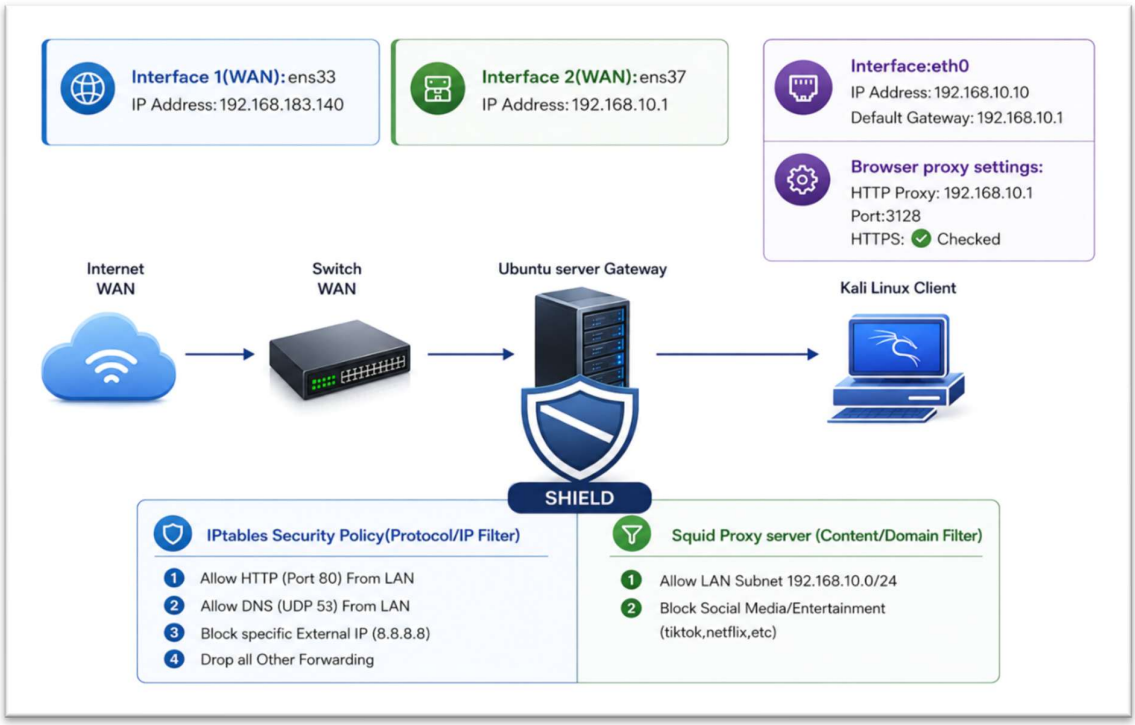
The physical router centers entirely on the Ubuntu Server gateway VM. To split our network zones cleanly, the Ubuntu gateway relies on two separate network adapters:

1. *ens33 (Interface 1 - WAN)*: Tied directly to the external virtual infrastructure to fetch an internet IP automatically via DHCP, resolving here to **192.168.183.140**.
2. *ens37 (Interface 2 - LAN)*: Manually forced up with a static IP assignment of **192.168.10.1** to act as our explicit private local gateway switch interface.

On our client machine running Kali Linux, the primary ethernet adapter (eth0) is assigned a static IP configuration of 192.168.10.10 pointing directly toward our Ubuntu interface (192.168.10.1) as its default gateway out. To channel web requests through our application layer protection, the Kali Linux client's browser proxy settings are explicitly configured to forward HTTP/HTTPS traffic to the gateway IP on port 3128.

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VIRTUAL NETWORK TOPOLOGY DIAGRAM



IP ADDRESSING TABLE

IP ADDRESSING TABLE

Device / Interface	Interface Name	IP Address	Subnet Mask	Default Gateway	Role
Ubuntu Gateway (WAN)	ens33	192.168.183.140	255.255.255.0	192.168.183.1*	Internet-facing Interface
Ubuntu Gateway (LAN)	ens37	192.168.10.1	255.255.255.0	N/A	LAN Gateway / Proxy Server
Kali Linux Client	eth0	192.168.10.10	255.255.255.0	192.168.10.1	Client Workstation

PROXY CONFIGURATION

Setting	Value
Proxy Server	192.168.10.1
Proxy Port	3128
HTTPS Proxy	Enabled

NETWORK SUMMARY

Network	Address Range	Purpose
WAN Network	192.168.183.0/24	Internet Connectivity
LAN Network	192.168.10.0/24	Internal Client Network

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2.2 Implementation setup

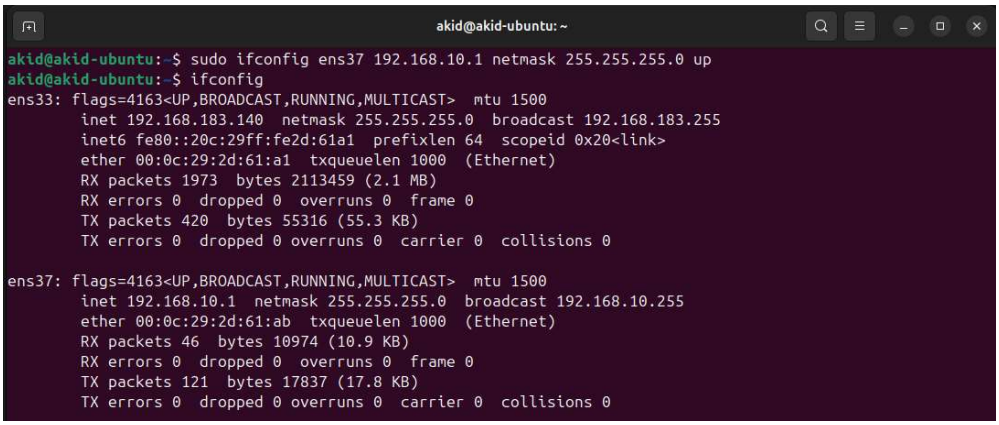
First, we will need to install Squid.

Step 1: Network configuration (The Foundation)

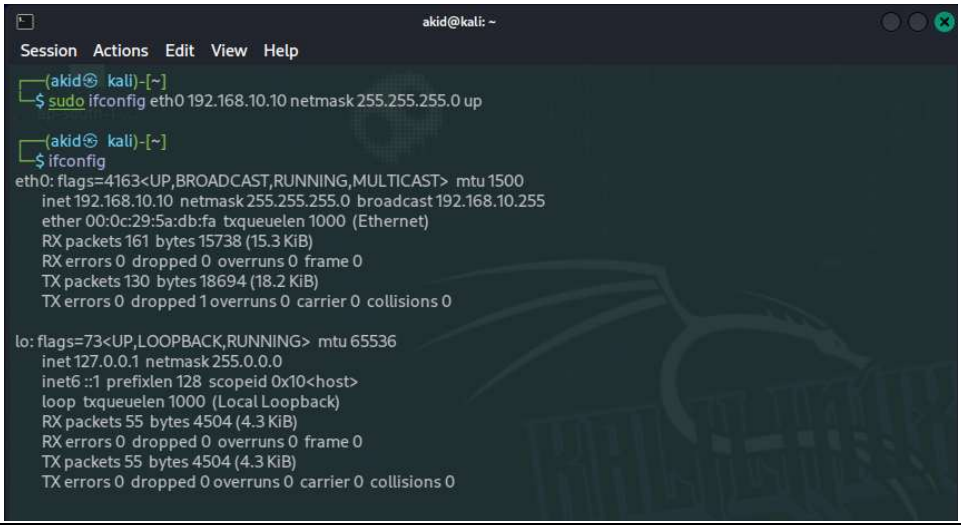
```
akid@akid-ubuntu:~$ sudo apt install squid -y
```

You must establish the IP addressing scheme for the "Internal LAN" as documented in your design.

1. On Ubuntu (The Gateway): Run this to assign the gateway IP to the interface connected to your LAN segment:



2. On Kali (The Client): Run these to give Kali an IP and point it toward the Ubuntu gateway:



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3. Verification: Try ping 192.168.10.1 from Kali. If it replies, your virtual "Internal LAN" is working.



Step 2: Security Policy Implementation

1. Persistent Routing and NAT

This ensures Kali can reach the internet through Ubuntu's ens33 interface.

```
akid@akid-ubuntu:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
akid@akid-ubuntu:~$ sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
```

2. Iptables Protocol Filtering

The project requires allowing only HTTP from the LAN.

- Allow Kali (ens37) to talk to the Squid service on Ubuntu

```
akid@akid-ubuntu:~$ sudo iptables -A INPUT -i ens37 -p tcp --dport 3128 -j ACCEPT
```

- Clear any old rules [cite: 14]

```
akid@akid-ubuntu:~$ sudo iptables -F
akid@akid-ubuntu:~$ sudo iptables -t nat -F
```

- Allow DNS (Port 53) - Vital for browsing to work

```
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -p udp --dport 53 -j ACCEPT
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -p tcp --dport 53 -j ACCEPT
```

- Allow only HTTP (Port 80) and HTTPS (Port 443) from LAN

```
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -i ens37 -p tcp --dport 80 -j ACCEPT
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -i ens37 -p tcp --dport 443 -j ACCEPT
```

- Block a specific external IP (common IP like 8.8.4.4)

```
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -d 8.8.4.4 -j DROP
```

- Drop all other traffic (Enforces the policy)

```
akid@akid-ubuntu:~$ sudo iptables -A FORWARD -j DROP
```


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```
akid@akid-ubuntu:~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 6619 packets, 7097K bytes)
 pkts bytes target     prot opt in     out     source               destination
 3899  577K ACCEPT     tcp  --  ens37  any     anywhere             anywhere             tcp dpt:3128
    0     0 ACCEPT     tcp  --  ens37  any     anywhere             anywhere             tcp dpt:3128

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     udp  --  any    any     anywhere             anywhere             udp dpt:domain
    0     0 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:domain
    0     0 ACCEPT     tcp  --  ens37  any     anywhere             anywhere             tcp dpt:http
    0     0 ACCEPT     tcp  --  ens37  any     anywhere             anywhere             tcp dpt:https
    0     0 DROP       all  --  any    any     anywhere             dns.google
    0     0 DROP       all  --  any    any     anywhere             anywhere

Chain OUTPUT (policy ACCEPT 3476 packets, 7765K bytes)
 pkts bytes target     prot opt in     out     source               destination
  587 47811 ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:http
 1985  363K ACCEPT     tcp  --  any    any     anywhere             anywhere             tcp dpt:https
  170 11854 ACCEPT     udp  --  any    any     anywhere             anywhere             udp dpt:domain
```

Step 3: Configuring Squid Proxy (Content Filter)

Now, we implement the Squid Proxy to block social media and entertainment.

1. **Edit the configuration: sudo nano /etc/squid/squid.conf.**

2. **Add these rules at the top of the file:**

```
# --- PROJECT CBS 2413 CONFIG START ---
# Define the internal LAN segment
acl internal_lan src 192.168.10.0/24

# Define the DNS server to use
dns_nameservers 8.8.8.8 8.8.4.4

# Security Policy: Block social media and entertainment websites
acl blocked_sites dstdomain .netflix.com .tiktok.com

# Apply access controls
http_access deny blocked_sites
http_access allow internal_lan
# --- PROJECT CBS 2413 CONFIG END ---
```

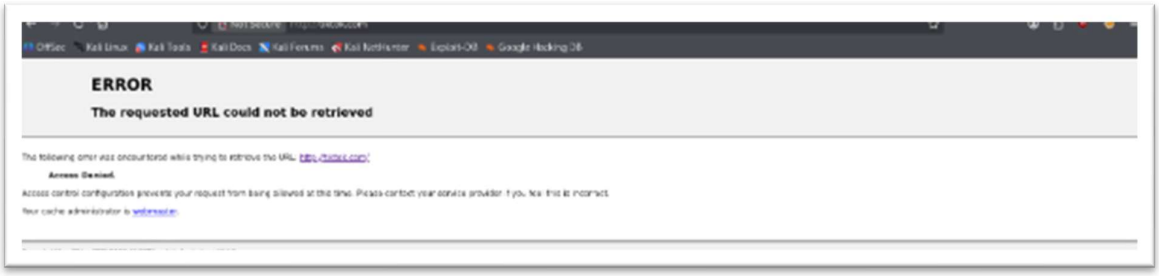
Step 4: Testing and Verification

- Netflix: Access denied



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- tiktok: denied access



- NeverSSL: allow access



- gmi vlu: allow access



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3.0 CONCLUSION

This configuration process shows how easily standard virtual machines can be configured into effective network edge security systems. By using iptables to manage layer-3/4 interfaces and routing structures, alongside Squid for context-aware application controls, we built a layered firewall deployment well-suited for a standard small business network.

The current setup relies on manual client-side configuration, which can be bypassed if users remove the proxy settings from their browsers. To improve this setup, future expansions should implement a Transparent Proxy Mode. This adjustment would use iptables PREROUTING rules to automatically intercept all outbound port 80/443 traffic at the gateway level, directing requests through Squid without requiring any manual device adjustments on the client workstations.

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4.0 REFERENCES

1) *Linux firewalls. (2025, August 6). No Starch Press. https://nostarch.com/firewalls.htm*
2) *Wessels, D. (2004, January 22). Squid: The Definitive Guide. O'Reilly Online Learning. https://www.oreilly.com/library/view/squid-the-definitive/0596001622/*
3) *What is a firewall? Definition and types of firewalls | Fortinet. (n.d.). Fortinet. https://www.fortinet.com/resources/cyberglossary/firewall*
4) *Netfilter Project. "Netfilter Project Documentation." netfilter.org. https://www.netfilter.org (accessed Jun. 6, 2026).*
5) *Final. (2018, April 29). How to setup a static IP on Ubuntu Server 18.04. Ask Ubuntu. https://askubuntu.com/questions/1029531/how-to-setup-a-static-ip-on-ubuntu-server-18-04*

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